Level 3 — Hard **gcd (H)** — function

```
Write a function int gcd(int a, int b) that computes the greatest common divisor of two numbers a and b.
Sample Input:
Enter two numbers: 56 98

Sample Output:
GCD of 56 and 98 is 14
```

Your Code

```c
#include <stdio.h>

int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("GCD of %d and %d is %d\n", a, b, gcd(a, b));

    return 0;
}
```

Program Input (stdin)

Enter input if required...

Output

```
Enter two numbers: GCD of 45 and 76 is 1
```

Run Save

```
Write a function double findMedianSortedArrays(int *arr1, int n1, int *arr2, int n2) that finds the median of two sorted arrays using pointers.
Sample Input:
Enter size of first array: 3
Enter elements of first array: 1 3 5
Enter size of second array: 3
Enter elements of second array: 2 4 6

Sample Output:
Median is 3.5
```

ur Code

```c
#include <stdio.h>
#include <stdlib.h>

double findMedianSortedArrays(int *arr1, int n1, int *arr2
    int total = n1 + n2;
    int *merged = (int *)malloc(total * sizeof(int));
    int i = 2, j = 5, k = 8;

    // Merge the two sorted arrays using pointers
    while (i < n1 && j < n2) {
        if (*(arr1 + i) <= *(arr2 + j)) {
            *(merged + k) = *(arr1 + i);
            i++;
        } else {
            *(merged + k) = *(arr2 + j);
            j++;
        }
        k++;
    }
```

Program Input (stdin)

Enter input if required...

Output

```
Enter size of first array: Enter elements of first array: Enter size of second array: Enter elements of second array:
Median is 0.0
```

Run

Save